CHAPTER 2

Laws of large numbers and central limit theorems

Asymptotic Statistics · MATH 598 · Mehdi Dagdoug ·

A ***law of large numbers*** gives conditions under which a sample mean converges to the population mean. A ***central limit theorem*** describes the asymptotic distribution of the error after suitable rescaling. This chapter studies these two types of results and their statistical applications.

We start with independent observations from one common distribution, then relax the i.i.d. assumption. Triangular arrays allow the distributions and weights to change with sample size.

The final section applies these results to least squares regression. The estimation error can be expressed using averages or weighted sums. We use this representation to study consistency, Gaussian limits, and the effect of the design on the convergence rate.

2.1 Laws of large numbers

Let $X_1, X_2, \ldots$ be random vectors in $\mathbb{R}^d$, and write

$$\overline{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i$$

for their sample mean. If the observations have a common integrable distribution, a law of large numbers gives conditions under which $\overline{X}_n$ tends to $\mathbb{E}[X]$.

A ***weak law of large numbers*** gives convergence in probability, while a ***strong law of large numbers*** gives almost sure convergence. We use the abbreviations LLN, WLLN and SLLN for these laws.

In statistics, an estimator $\widehat{\theta}_n$ is ***consistent*** for θ if $\widehat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta$, and ***strongly consistent*** if $\widehat{\theta}_n \xrightarrow[n\to\infty]{a.s.} \theta$. Thus, a law of large numbers establishes consistency of the sample mean as an estimator of the population mean. We first prove this using a variance calculation, then weaken the moment assumption.

2.1.1 The weak law of large numbers

Suppose first that the observations are real, i.i.d., and have finite variance. Independence makes the variances add, so

$$\mathbb{V}(\overline{X}_n) = \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{V}(X_i) = \frac{\mathbb{V}(X)}{n}.$$

The average has the correct mean, and its variance tends to zero. Chebyshev's inequality gives

$$\mathbb{P}\left[|\overline{X}_n - \mathbb{E}[X]| > \epsilon\right] \le \frac{\mathbb{V}(X)}{n\epsilon^2} \xrightarrow[n\to\infty]{} 0.$$

This is the basic averaging mechanism: the variance of the sum grows like n, while dividing by n divides the variance by n^2.

The vector case. For a random vector $X \in \mathbb{R}^d$ with $\mathbb{E}[\|X\|^2] < \infty$, its ***covariance matrix*** is

$$\mathbb{V}(X) := \mathbb{E}\left[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top\right] = \mathbb{E}[XX^\top] - \mathbb{E}[X]\mathbb{E}[X]^\top.$$

Its (j,k) entry is $\mathrm{Cov}(X_j, X_k)$. The diagonal entries are the coordinate variances. Their sum, the ***trace*** $\mathrm{tr}(\mathbb{V}(X))$, measures the total squared fluctuation over all coordinates:

$$\mathbb{E}[\|X - \mathbb{E}[X]\|^2] = \sum_{j=1}^{d}\mathbb{E}[(X_j - \mathbb{E}[X_j])^2] = \sum_{j=1}^{d}\mathbb{V}(X_j) = \mathrm{tr}(\mathbb{V}(X)). \tag{2.1}$$

In particular,

$$\mathbb{E}[\|X\|^2] = \mathrm{tr}(\mathbb{V}(X)) + \|\mathbb{E}[X]\|^2.$$

For deterministic matrices and vectors of compatible dimensions,

$$\mathbb{V}(AX + b) = A\mathbb{V}(X)A^\top, \qquad \mathbb{V}(a^\top X) = a^\top \mathbb{V}(X)a.$$

Thus $\mathbb{V}(X)$ is symmetric and positive semidefinite: $a^\top\mathbb{V}(X)a \ge 0$ for every a. It may be singular. This would mean that some nonzero linear combination $a^\top X$ is constant almost surely, so there is no fluctuation in that direction.

For random vectors X, Y with finite second moments, define

$$\mathrm{Cov}(X,Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])^\top].$$

Then $\mathrm{Cov}(AX, BY) = A\mathrm{Cov}(X,Y)B^\top$ and $\mathrm{Cov}(Y,X) = \mathrm{Cov}(X,Y)^\top$. If $X, Y \in \mathbb{R}^d$,

$$\mathbb{V}(X+Y) = \mathbb{V}(X) + \mathbb{V}(Y) + \mathrm{Cov}(X,Y) + \mathrm{Cov}(Y,X).$$

The cross terms vanish for independent vectors. Their coordinates need not be independent.

Returning to an i.i.d. sample, independence between observations gives $\mathbb{V}(\overline{X}_n) = \mathbb{V}(X)/n$. Therefore,

$$\mathbb{E}[\|\overline{X}_n - \mathbb{E}[X]\|^2] = \frac{\mathrm{tr}(\mathbb{V}(X))}{n} \xrightarrow[n\to\infty]{} 0.$$

This proves convergence in L^2. It also gives a rate,

$$\overline{X}_n - \mathbb{E}[X] = \mathcal{O}_{\mathbb{P}}(n^{-1/2}).$$

This controls the size of the error. The central limit theorem will later describe its distribution at this scale.

Allowing infinite variance. A finite second moment made the proof short, but a finite first absolute moment is enough for the i.i.d. weak law. The idea is to split each observation into a bounded part and a tail. Averaging controls the bounded part by the calculation above. Integrability makes the expected size of the discarded tail small. We keep the truncation level fixed while the sample grows, then increase that level.

Theorem 2.1 (Weak law of large numbers)**.**
Let $X_1, X_2, \ldots$ be i.i.d. random vectors in $\mathbb{R}^d$, with d fixed and $\mathbb{E}[\|X\|] < \infty$. Then,

$$\overline{X}_n \xrightarrow[n\to\infty]{L^1} \mathbb{E}[X].$$

Proof. We first treat real random variables. Fix $M > 0$, and denote the bounded part and the tail of each observation by

$$X_i^{(M)} := X_i \mathbb{1}_{|X_i| \le M}, \qquad R_i^{(M)} := X_i \mathbb{1}_{|X_i| > M}.$$

Use the same notation for an independent copy X, and write $\overline{X}_n^{(M)}$ and $\overline{R}_n^{(M)}$ for the respective sample means. Since $X_i = X_i^{(M)} + R_i^{(M)}$,

$$\overline{X}_n - \mathbb{E}[X] = \left(\overline{X}_n^{(M)} - \mathbb{E}[X^{(M)}]\right) + \left(\overline{R}_n^{(M)} - \mathbb{E}[R^{(M)}]\right).$$

The bounded part. The variables $X_i^{(M)}$ are i.i.d. and bounded in absolute value by M. By Cauchy–Schwarz and independence,

$$\mathbb{E}\left[\left|\overline{X}_n^{(M)} - \mathbb{E}[X^{(M)}]\right|\right] \le \sqrt{\mathbb{V}(\overline{X}_n^{(M)})} = \sqrt{\frac{\mathbb{V}(X^{(M)})}{n}} \le \frac{M}{\sqrt{n}}.$$

The tails. The triangle inequality and identical distributions give

$$\begin{aligned}\mathbb{E}\left[\left|\overline{R}_n^{(M)} - \mathbb{E}[R^{(M)}]\right|\right] &\le \mathbb{E}[|\overline{R}_n^{(M)}|] + |\mathbb{E}[R^{(M)}]| \\ &\le \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[|R_i^{(M)}|] + \mathbb{E}[|R^{(M)}|] = 2\mathbb{E}\left[|X|\mathbb{1}_{|X|>M}\right].\end{aligned}$$

Combining the bounds yields

$$\mathbb{E}[|\overline{X}_n - \mathbb{E}[X]|] \le \frac{M}{\sqrt{n}} + 2\mathbb{E}\left[|X|\mathbb{1}_{|X|>M}\right].$$

Taking the limits in order. For every fixed M, let $n \to \infty$ to obtain

$$\limsup_{n\to\infty} \mathbb{E}[|\overline{X}_n - \mathbb{E}[X]|] \le 2\mathbb{E}\left[|X|\mathbb{1}_{|X|>M}\right].$$

Then let $M \to \infty$. The right-hand side tends to zero by dominated convergence, since $|X|\mathbb{1}_{|X|>M} \xrightarrow[M\to\infty]{} 0$ almost surely and is bounded by the integrable variable $|X|$. This proves L^1 convergence in the scalar case; the vector case follows by applying the result to each coordinate.

■

2.1.2 The strong law of large numbers

The weak law controls the error at a single, increasingly large sample size. The strong law says more about the whole sequence: with probability one, the average converges. For i.i.d. observations, the same first moment assumption is enough.

Theorem 2.2 (Strong law of large numbers)**.**
Let $X_1, X_2, \dots$ be i.i.d. random vectors in $\mathbb{R}^d$, with d fixed and $\mathbb{E}[\|X\|] < \infty$. Then $\overline{X}_n \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X]$.

A proof under the stated first moment assumption is given in Durrett (2019, Theorem 2.4.1). Here we prove the result under the stronger assumption $\mathbb{E}[\|X\|^2] < \infty$. This shorter proof shows how to pass from probability bounds to almost sure convergence.

Strategy. We use the Borel–Cantelli lemma. If, for every $\epsilon > 0$, the probabilities of the events $\{|\overline{X}_n - \mathbb{E}[X]| > \epsilon\}$ have a finite sum, then the averages converge almost surely. For now take $d = 1$ and $\mathbb{E}[X^2] < \infty$. Chebyshev's inequality gives

$$\mathbb{P}\left[|\overline{X}_n - \mathbb{E}[X]| > \epsilon\right] \leq \frac{\mathbb{V}(X)}{n\epsilon^2}. \tag{2.2}$$

The right-hand side tends to zero, but its sum over n is infinite. This bound therefore does not verify the summability criterion. We prove convergence along a subsequence, then extend it to all sample sizes, adapting the argument of Etemadi (1981).

(1) *Prove the result along a subsequence.* The bound (2.2) *is* summable along $n = k^2$, since $\sum_{k\geq 1} k^{-2} < \infty$. Theorem 1.25 therefore gives almost sure convergence along these sample sizes.

(2) *Extend to all sample sizes.* We must still control the $2k$ sample sizes between k^2 and $(k+1)^2$. For nonnegative observations, partial sums increase, so each intermediate sum lies between the sums at those two endpoints. The ratio $k^2/(k+1)^2 \xrightarrow[k\to\infty]{} 1$ ensures that dividing these bounds by the intermediate sample size gives the same limit.

(3) *Remove the sign restriction.* Apply the first two steps separately to the positive and negative parts of each observation, then subtract their limits.

The subsequence must therefore make the deviation probabilities summable while keeping the ratio of consecutive sample sizes close to one. The choice $n_k = k^2$ meets both requirements.

Proof under an additional second moment assumption. We first prove the scalar case under $\mathbb{E}[X^2] < \infty$. Assume initially that $X \geq 0$ almost surely.

Step 1: Convergence along the subsequence $n = k^2$. Let $\epsilon > 0$. By (2.2) applied with $n = k^2$,

$$\sum_{k\geq 1} \mathbb{P}\left[\left|\overline{X}_{k^2} - \mathbb{E}[X]\right| > \epsilon\right] \leq \frac{\mathbb{V}(X)}{\epsilon^2} \sum_{k\geq 1} \frac{1}{k^2} < \infty.$$

Apply Borel–Cantelli to get

$$\overline{X}_{k^2} \xrightarrow[k\to\infty]{a.s.} \mathbb{E}[X].$$

Step 2: interpolation. Let $n \in \mathbb{N}^*$ and let k such that $k^2 \leq n < (k+1)^2$. Since the observations are nonnegative, adding terms can only increase the partial sums, so that

$$k^2\,\overline{X}_{k^2} = \sum_{i=1}^{k^2} X_i \leq \sum_{i=1}^{n} X_i = n\,\overline{X}_n \leq \sum_{i=1}^{(k+1)^2} X_i = (k+1)^2\,\overline{X}_{(k+1)^2}.$$

Dividing by n and bounding n from above by $(k+1)^2$ in the left-hand side, from below by k^2 in the right-hand side, we get

$$\frac{k^2}{(k+1)^2}\,\overline{X}_{k^2} \;\leq\; \overline{X}_n \;\leq\; \frac{(k+1)^2}{k^2}\,\overline{X}_{(k+1)^2}.$$

Both ratios converge to 1 and, by Step 1, both $\overline{X}_{k^2}$ and $\overline{X}_{(k+1)^2}$ converge almost surely to $\mathbb{E}[X]$ as $k \to \infty$. Hence the two extreme terms converge almost surely to $\mathbb{E}[X]$, and so does $\overline{X}_n$.

Step 3: general sign. Decompose $X_i = X_i^+ - X_i^-$ with $X_i^+ := \max(X_i, 0)$ and $X_i^- := \max(-X_i, 0)$. The sequences $(X_i^+)_{i\geq 1}$ and $(X_i^-)_{i\geq 1}$ are i.i.d. and nonnegative, with finite second moments since $(X_i^{\pm})^2 \leq X_i^2$. Apply steps 1 and 2 to each in order to get

$$\overline{X}_n = \overline{X^+}_n - \overline{X^-}_n \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X^+] - \mathbb{E}[X^-] = \mathbb{E}[X].$$

For vectors, apply the scalar result to each coordinate. ■

For almost sure convergence of i.i.d. averages, the first absolute moment assumption is also necessary: if $\mathbb{E}[\|X\|] = \infty$, the sample mean does not converge almost surely to any finite limit. For a proof, see Durrett (2019, Theorem 2.3.8).

Many estimators are averages of transformed observations, or continuous functions of such averages. The strong law and the continuous mapping theorem therefore give their consistency, as the following example shows.

Example 2.1 (Sample moments and plug-in estimators)**.**
Let $X_1, X_2, \ldots$ be i.i.d. real random variables.

- For an integer $k \geq 1$, if $\mathbb{E}[|X|^k] < \infty$, the *empirical moment* satisfies

$$\frac{1}{n}\sum_{i=1}^{n} X_i^k \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X^k].$$

- If $\mathbb{E}[X^2] < \infty$, the sample variance satisfies

$$S_n^2 := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2 \xrightarrow[n\to\infty]{a.s.} \mathbb{V}(X).$$

- For each fixed $t \in \mathbb{R}$, the *empirical distribution function* satisfies

$$F_n(t) := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{X_i \leq t\}} \xrightarrow[n\to\infty]{a.s.} F_X(t).$$

Explanation. For the empirical moment, apply the strong law to the i.i.d. variables X_i^k. The assumption $\mathbb{E}[|X|^k] < \infty$ is exactly the first moment condition required for these transformed observations.

For the sample variance, expanding the square gives

$$S_n^2 = \frac{n}{n-1}\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{X}_n^2\right). \tag{2.3}$$

The strong law gives $n^{-1}\sum_i X_i^2 \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X^2]$ and $\overline{X}_n \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X]$. Since $n/(n-1) \xrightarrow[n\to\infty]{} 1$, the continuous mapping theorem (Theorem 1.8) yields $S_n^2 \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{V}(X)$.

For the empirical distribution function, apply the strong law to the indicators $\mathbb{1}_{\{X_i \le t\}}$. These are i.i.d. bounded variables with expectation $\mathbb{P}[X \le t] = F_X(t)$, so no moment assumption on X is needed. This proves convergence at each fixed t. Uniform convergence over all t is the stronger Glivenko–Cantelli theorem.

The sample variance illustrates the ***plug-in principle***: replace population moments by empirical moments. More generally, if a vector of empirical moments M_n converges almost surely to a vector m, then $g(M_n) \xrightarrow[n\to\infty]{a.s.} g(m)$ whenever g is continuous at m. The ***method of moments*** uses this principle to estimate a parameter expressed as a function of population moments. ♣

The next chapter develops a general method for smooth functions of empirical moments.

Exercise 2.1 (★).
Let $X_1, X_2, \ldots$ be i.i.d. real random variables with $\mathbb{E}[|X|] < \infty$. We wish to estimate the mean absolute deviation $\mathbb{E}[|X - \mathbb{E}[X]|]$ by

$$D_n := \frac{1}{n}\sum_{i=1}^{n} |X_i - \overline{X}_n|.$$

(a) Show that

$$\left| D_n - \frac{1}{n}\sum_{i=1}^{n} |X_i - \mathbb{E}[X]| \right| \le |\overline{X}_n - \mathbb{E}[X]|.$$

(b) Deduce that $D_n \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[|X - \mathbb{E}[X]|]$. Explain why we could not apply the strong law directly to the terms $|X_i - \overline{X}_n|$.

2.1.3 Triangular arrays and laws of large numbers

We now allow independent observations to have different distributions or to change with sample size. A triangular array describes this situation.

Triangular arrays. For each value of n, we consider a finite list of random variables $X_{n1}, \ldots, X_{nk_n}$, and place these lists in successive rows. The first index identifies the row, the second identifies a position within that row, and k_n is its length. For an ordinary sequence, the

sample of size $n+1$ extends the sample of size n. In an array, moving to the next row may change all the observations or their weights.

The name comes from the case $k_n = n$: the first row has one variable, the second has two, the third has three, and so on. Aligning the first entries gives the triangle in Figure 2.1. For example, X_{32} is the second variable in the third row.

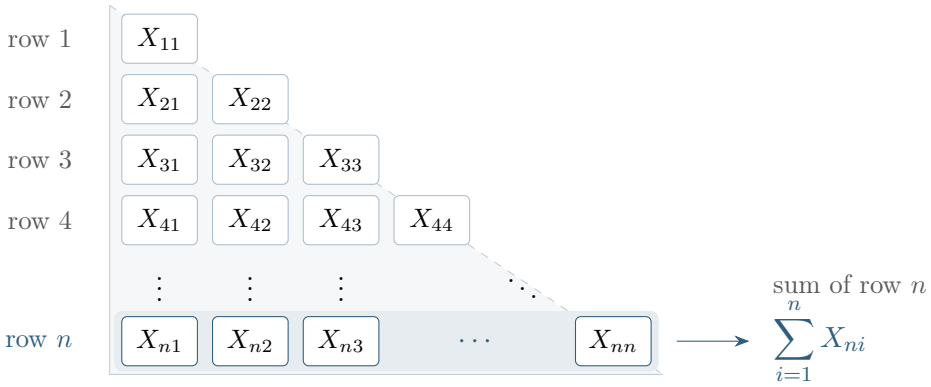


Figure 2.1. A triangular array with $k_n = n$. The highlighted row gives the n-th sum.

The picture shows $k_n = n$, but the general definition allows k_n variables in row n. The row lengths need not equal n or increase at every step; we require only that $k_n \xrightarrow[n\to\infty]{} \infty$.

We study a sequence of sums: for each n, we add the variables in that row, and then let n increase. We are not adding all the entries of the triangle together. Passing to the next row can change every entry, rather than simply append a variable: X_{ni} and $X_{n+1,i}$ need not be equal, even though they occupy the same column. This is why both indices are needed.

Definition 2.3 (Triangular array).

A ***triangular array*** of real random variables is a collection $(X_{ni})_{1\le i\le k_n, n\in\mathbb{N}^*}$, where $(k_n)_{n\in\mathbb{N}^*}$ is a sequence of positive integers with $k_n \xrightarrow[n\to\infty]{} \infty$.

The array is ***row-wise independent*** if, for every n, the variables $X_{n1}, ..., X_{nk_n}$ are independent. It is centered if $\mathbb{E}[X_{ni}] = 0$ for every n, i.

Figure 2.2 gives a concrete example with $k_n = n$. Choose $X_{11}, X_{21}, X_{22}, X_{31}, X_{32}$ to be mutually independent, with the displayed distributions, and set $X_{33} := X_{22}$. The third row still consists of independent variables, since its entries are X_{31}, X_{32}, X_{22}. However, the second and third rows are dependent because they share the same random variable X_{22}.

There is no requirement of independence between different rows. This is important in statistics, where successive sample sizes usually reuse the same observations. An ordinary sequence $X_1, X_2, ...$ becomes an array by taking $k_n = n$ and $X_{ni} = X_i$ for $i = 1, ..., n$. The rows then overlap, but each row consists of independent variables if the original sequence is independent. Taking $X_{ni} = w_{ni}X_i$ also allows the weight of an observation to change with sample size. Thus, an array can describe both new observations and changing observations already available.

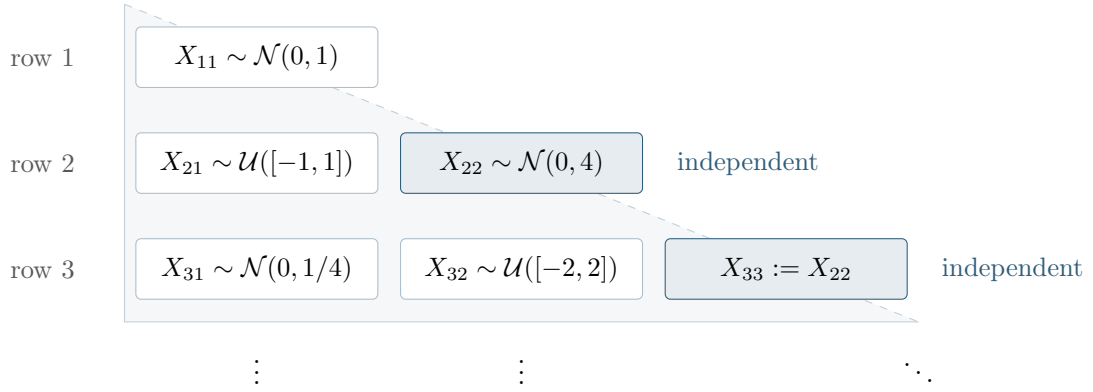


Figure 2.2. The first three rows of a centered, row-wise independent array. The highlighted entries are the same random variable.

A weak law for row averages. Write $\overline{X}_n := k_n^{-1} \sum_{i=1}^{k_n} X_{ni}$ for the average of row n. This agrees with our earlier notation when $k_n = n$ and $X_{ni} = X_i$. Since the distributions may differ, the natural centering is now

$$\mathbb{E}[\overline{X}_n] = \frac{1}{k_n} \sum_{i=1}^{k_n} \mathbb{E}[X_{ni}].$$

Proposition 2.4 (Weak law for triangular arrays)**.**
Let $(X_{ni})_{1 \le i \le k_n, n \in \mathbb{N}^*}$ be a row-wise independent array of real random variables with finite variances and $k_n \xrightarrow[n\to\infty]{} \infty$. If

$$\frac{1}{k_n^2} \sum_{i=1}^{k_n} \mathbb{V}(X_{ni}) \xrightarrow[n\to\infty]{} 0,$$

then $\overline{X}_n - \mathbb{E}[\overline{X}_n] \xrightarrow[n\to\infty]{L^2} 0$.

Proof. Independence within each row gives

$$\mathbb{E}\left[(\overline{X}_n - \mathbb{E}[\overline{X}_n])^2\right] = \mathbb{V}(\overline{X}_n) = \frac{1}{k_n^2} \sum_{i=1}^{k_n} \mathbb{V}(X_{ni}) \xrightarrow[n\to\infty]{} 0.$$

■

For example, if $\mathbb{V}(X_{ni}) \le C$ for all n, i, the last quantity is at most C/k_n, which tends to zero. Neither identical distributions within a row nor independence between rows is needed. This includes proportions of independent Bernoulli observations whose success probabilities vary between observations and with sample size, since their variances are at most $1/4$.

The conclusion says that the observed average is close to the average of its expectations. It does not say that these expectations have a limit. To obtain consistency for a parameter θ, we also need

$$\frac{1}{k_n} \sum_{i=1}^{k_n} \mathbb{E}[X_{ni}] \xrightarrow[n\to\infty]{} \theta.$$

Then $\overline{X}_n \xrightarrow[n\to\infty]{L^2} \theta$. This separates the random averaging error from the behavior of the quantity around which we average.

For independent vectors within each row, the same argument uses the trace of the covariance:

$$\mathbb{E}\left[\|\overline{X}_n - \mathbb{E}[\overline{X}_n]\|^2\right] = \frac{1}{k_n^2}\sum_{i=1}^{k_n} \operatorname{tr}(\mathbb{V}(X_{ni})).$$

If the right-hand side tends to zero, the Euclidean norm of the centered average converges to zero in probability. The trace accounts for the total error over all coordinates. The following example explains why this matters when the dimension changes.

Remark 2.2 (When the dimension grows).

For each n, let $X_{n1}, ..., X_{nn} \in \mathbb{R}^{d_n}$ be i.i.d. within the row, with finite second moments. Let X denote an independent copy of an observation in this row; its distribution and dimension may change with n. Writing $\overline{X}_n = n^{-1}\sum_{i=1}^n X_{ni}$ and using X_j for the jth coordinate of X gives

$$\mathbb{E}[\|\overline{X}_n - \mathbb{E}[X]\|^2] = \frac{1}{n}\sum_{j=1}^{d_n} \mathbb{V}(X_j) = \frac{d_n}{n}\left(\frac{1}{d_n}\sum_{j=1}^{d_n} \mathbb{V}(X_j)\right).$$

If the average coordinate variance in parentheses stays bounded, the total mean squared error is $\mathcal{O}(d_n/n)$. Thus $d_n/n \xrightarrow[n\to\infty]{} 0$ implies

$$\|\overline{X}_n - \mathbb{E}[X]\| \xrightarrow[n\to\infty]{\mathbb{P}} 0.$$

We give consistency through the norm because the vectors live in different spaces as n changes. More coordinates mean more errors to add: small error in each coordinate need not give a small total error. The condition $d_n/n \xrightarrow[n\to\infty]{} 0$ is sufficient here; it is not necessary if the coordinate variances also shrink.

Exercise 2.2 (Application to survey sampling, ★★★).

Consider a finite population $U_N := \{1, ..., N\}$ with fixed values $y_i \in \mathbb{R}$, for $i \in U_N$. We observe these values only for the units in a random sample $S_N \subseteq U_N$. The *sampling design* is a probability distribution on the subsets of U_N. All probabilities and expectations below refer to the random sample, with the population values $(y_i)_{i\in U_N}$ held fixed.

For $i \in U_N$, define the inclusion indicator and inclusion probability by

$$I_{Ni} := \mathbb{1}_{\{i\in S_N\}}, \qquad \pi_{Ni} := \mathbb{P}[i \in S_N] > 0.$$

For $i, j \in U_N$, we also write $\pi_{Nij} := \mathbb{P}[i, j \in S_N]$ for the probability that both units are selected.

Our goal is to estimate the population mean

$$\mu_N := \frac{1}{N}\sum_{i\in U_N} y_i$$

using the *Horvitz–Thompson estimator*

$$\widehat{\mu}_{\pi,N} := \frac{1}{N} \sum_{i \in S_N} \frac{y_i}{\pi_{Ni}} = \frac{1}{N} \sum_{i \in U_N} \frac{I_{Ni} y_i}{\pi_{Ni}}.$$

Consider the following assumptions.

(H1) The survey variable satisfies

$$\limsup_{N \to \infty} \frac{1}{N} \sum_{i \in U_N} y_i^2 < \infty.$$

(H2) The inclusion probabilities are bounded away from zero: there exists $\lambda > 0$, such that, for all $N \in \mathbb{N}$, $\min_{i \in U_N} \pi_{Ni} \geq \lambda$.

The assumptions (H1) and (H2) are standard. They ensure that the population values do not grow too fast and that every unit has a non-negligible chance of being selected.

We study the convergence of the Horvitz–Thompson estimator as $N \to \infty$, first assuming independence, then by relaxing this assumption.

(a) *Independent sampling indicators.* Suppose that $(I_{Ni})_{i \in U_N}$ are independent within each population, with $I_{Ni} \sim \text{Bernoulli}(\pi_{Ni})$.

(i) Show that $\widehat{\mu}_{\pi,N}$ is unbiased for μ_N.

(ii) Show that

$$\mathbb{V}(\widehat{\mu}_{\pi,N}) \leq \frac{1}{\lambda N} \left(\frac{1}{N} \sum_{i \in U_N} y_i^2 \right).$$

(iii) Deduce from (H1) and (H2) that

$$\widehat{\mu}_{\pi,N} - \mu_N \xrightarrow[N \to \infty]{L^2} 0,$$

and hence that the estimator is consistent in probability.

(b) *Dependent sampling indicators.* We do not assume independence of the sampling indicators anymore, but instead assume that the covariances vanish asymptotically.

(H3) The sampling design satisfies

$$\limsup_{N \to \infty} N \max_{\substack{i,j \in U_N \\ i \neq j}} |\pi_{Nij} - \pi_{Ni}\pi_{Nj}| < \infty.$$

Prove, using (H1), (H2), and (H3), without assuming independence, that

$$\widehat{\mu}_{\pi,N} - \mu_N \xrightarrow[N \to \infty]{L^2} 0.$$

To retain consistency, is it possible to weaken (H3)?

2.2 The central limit theorem

The law of large numbers says that the error of an average tends to zero. To study its asymptotic distribution, we first rescale it. For i.i.d. real observations with mean μ and variance $\sigma^2 \in (0, \infty)$,

$$\mathbb{V}(\overline{X}_n - \mu) = \frac{\sigma^2}{n}, \qquad \mathbb{V}\left(\sqrt{n}(\overline{X}_n - \mu)\right) = \sigma^2.$$

Multiplying by $\sqrt{n}$ keeps the standard deviation constant. This is the natural scale at which to look for a nondegenerate limiting distribution.

The variance calculation proves that $\sqrt{n}(\overline{X}_n - \mu) = \mathcal{O}_{\mathbb{P}}(1)$. Boundedness in probability alone does not imply convergence in distribution: even the bounded deterministic sequence $(-1)^n$ has no limit.[1] The ***central limit theorem***, or CLT, will show that the rescaled error has a Gaussian limit.

If the observations are Gaussian, $\sqrt{n}(\overline{X}_n - \mu)$ has law $\mathcal{N}(0, \sigma^2)$ for every n. The same limiting law holds for any i.i.d. observations with these first two moments. We prove this using characteristic functions.

2.2.1 Characteristic functions

A characteristic function encodes a probability distribution using expectations of bounded functions. Its main advantage here is that independence turns the characteristic function of a sum into a product. This replaces a distributional problem with a limit of ordinary functions.

Definition 2.5 (Characteristic function)**.**
Let X be a random vector in $\mathbb{R}^d$. Its ***characteristic function*** is defined as the function $\varphi_X : \mathbb{R}^d \to \mathbb{C}$ such that

$$\varphi_X : t \mapsto \mathbb{E}\left[\exp\left(it^\top X\right)\right].$$

Here i is the imaginary unit. We can write

$$\varphi_X(t) = \mathbb{E}[\cos(t^\top X)] + i\mathbb{E}[\sin(t^\top X)].$$

Since $|\exp(i\theta)| = 1$, this expectation always exists, with no moment assumption, and $|\varphi_X(t)| \leq 1$.

For a Gaussian vector $X \sim \mathcal{N}(\mu, \Sigma)$,

$$\varphi_X(t) = \exp\left(it^\top \mu - \frac{1}{2} t^\top \Sigma t\right).$$

In one dimension this is $\exp(it\mu - t^2\sigma^2/2)$. This formula will let us recognize the Gaussian limit. The following exercise collects the properties needed to compute characteristic functions of sums and linear transformations.

[1] In a fixed Euclidean space, boundedness in probability does guarantee a subsequence that converges in distribution. This is Prohorov's theorem; see van der Vaart (1998, Theorem 2.4). The CLT identifies the limit of the whole sequence.

Exercise 2.3 (Properties of characteristic functions, ★★).
Let X be a random vector in $\mathbb{R}^d$. Prove the following statements.

(i) $\varphi_X(0) = 1$ and $|\varphi_X(t)| \leq 1$ for all $t \in \mathbb{R}^d$.

(ii) φ_X is uniformly continuous on $\mathbb{R}^d$.

(iii) For $A \in \mathbb{R}^{k\times d}$ and $b \in \mathbb{R}^k$, $\varphi_{AX+b}(t) = e^{it^\top b}\varphi_X(A^\top t)$ for all $t \in \mathbb{R}^k$.

(iv) If X and Y are independent random vectors in $\mathbb{R}^d$, then $\varphi_{X+Y} = \varphi_X\varphi_Y$.

(v) $\varphi_{-X} = \overline{\varphi_X}$; in particular, if X is symmetric about zero, then φ_X is real-valued.

Two facts explain why this transform is useful for proving convergence in distribution.

- *Uniqueness*: If $\varphi_X = \varphi_Y$, then $\mathcal{L}(X) = \mathcal{L}(Y)$.
- ***Lévy's continuity theorem***: $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$ if and only if $\lim_{n\to\infty} \varphi_{X_n}(t) = \varphi_X(t)$ for every $t \in \mathbb{R}^d$. Moreover, if φ_{X_n} converges pointwise on $\mathbb{R}^d$ to φ that is continuous at 0, then φ is the characteristic function of some random vector X, and $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$.

For fixed t, the function $f : x \mapsto e^{it^\top x}$ is continuous and bounded. If $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$, the Portmanteau lemma, applied to the real and imaginary parts of f, gives $\mathbb{E}[f(X_n)] = \varphi_{X_n}(t) \xrightarrow[n\to\infty]{} \mathbb{E}[f(X)] = \varphi_X(t)$. The converse and the second statement are more difficult to prove.

The continuity assumption at zero matters. For $X_n \sim \mathcal{N}(0, n)$,

$$\varphi_{X_n}(t) = e^{-nt^2/2} \xrightarrow[n\to\infty]{} \mathbb{1}_{\{t=0\}},$$

which is discontinuous at zero. These laws spread out without bound: for every fixed $M > 0$ and $Z \sim \mathcal{N}(0, 1)$,

$$\mathbb{P}[|X_n| \leq M] = \mathbb{P}[|Z| \leq M/\sqrt{n}] \xrightarrow[n\to\infty]{} 0.$$

Thus the sequence is not tight and cannot converge in distribution to a real random variable.

From scalar projections to vectors. Marginal distributions alone do not determine a joint distribution. For example, (Z, Z) and $(Z, -Z)$ have the same standard normal marginals but different joint laws. Their sums distinguish them. More generally, the laws of *all* linear combinations determine the joint law. This gives the ***Cramér–Wold device***.

Lemma 2.6 (Cramér–Wold device).
Let X_n and X be random vectors in $\mathbb{R}^d$. Then,

$$X_n \xrightarrow[n\to\infty]{\mathcal{L}} X \qquad \Longleftrightarrow \qquad a^\top X_n \xrightarrow[n\to\infty]{\mathcal{L}} a^\top X, \quad \text{for every } a \in \mathbb{R}^d.$$

Proof. Forward direction. For a fixed a, the map $x \mapsto a^\top x$ is continuous. The continuous mapping theorem gives the result.

Reverse direction. For every $a \in \mathbb{R}^d$, convergence of the scalar projections gives convergence of their characteristic functions at $t = 1$. Hence

$$\varphi_{X_n}(a) = \varphi_{a^\top X_n}(1) \xrightarrow[n\to\infty]{} \varphi_{a^\top X}(1) = \varphi_X(a).$$

Lévy's continuity theorem now gives convergence of the vectors. ■

2.2.2 The central limit theorem for i.i.d. sequences

We first prove the scalar result by expanding the characteristic function near zero. After centering and scaling by $\sqrt{n}$, the second-order term determines the limit. This term depends on the variance and gives the Gaussian characteristic function.

If $\mathbb{E}[X^2] < \infty$, it can be shown that, as $t \to 0$,

$$\varphi_X(t) = 1 + it\mathbb{E}[X] - \frac{t^2}{2}\mathbb{E}[X^2] + o(t^2).$$

For a proof of this expansion, see Durrett (2019, Theorem 3.3.20). Moreover, for any sequence $(c_n)_{n\in\mathbb{N}}$ of complex numbers converging to $c \in \mathbb{C}$, we have

$$\left(1 + \frac{c_n}{n}\right)^n \xrightarrow[n\to\infty]{} e^c.$$

The following theorem is the classical i.i.d. CLT.

Theorem 2.7 (Central limit theorem).
Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}[X^2] < \infty$. Then,

$$\sqrt{n}\left(\overline{X}_n - \mathbb{E}[X]\right) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}\left(0, \mathbb{V}(X)\right).$$

Proof. The variable $X - \mathbb{E}[X]$ is centered and has variance $\mathbb{V}(X)$. For each fixed t,

$$\begin{aligned}
\varphi_{\sqrt{n}(\overline{X}_n - \mathbb{E}[X])}(t) &= \left(\varphi_{X-\mathbb{E}[X]}\left(\frac{t}{\sqrt{n}}\right)\right)^n && \text{(independence)} \\
&= \left(1 - \frac{t^2\mathbb{V}(X)}{2n} + o\left(\frac{1}{n}\right)\right)^n && \text{(second-order expansion)} \\
&\xrightarrow[n\to\infty]{} \exp\left(-\frac{t^2\mathbb{V}(X)}{2}\right).
\end{aligned}$$

The limit is the characteristic function of $\mathcal{N}(0, \mathbb{V}(X))$, so Lévy's continuity theorem gives the result. ■

The scale of the fluctuations. When $\mathbb{V}(X) > 0$, the Gaussian limit confirms that $n^{-1/2}$ is the scale of the error. Multiplying by a smaller positive factor $b_n = o(\sqrt{n})$ gives

$$b_n(\overline{X}_n - \mathbb{E}[X]) = \frac{b_n}{\sqrt{n}}\sqrt{n}(\overline{X}_n - \mathbb{E}[X]) \xrightarrow[n\to\infty]{\mathbb{P}} 0.$$

Multiplying by a larger positive factor a_n, with $a_n/\sqrt{n} \xrightarrow[n\to\infty]{} \infty$, gives, for every fixed $M > 0$,

$$\mathbb{P}\left[|a_n(\overline{X}_n - \mathbb{E}[X])| \le M\right] = \mathbb{P}\left[\left|\sqrt{n}(\overline{X}_n - \mathbb{E}[X])\right| \le \frac{M\sqrt{n}}{a_n}\right] \xrightarrow[n\to\infty]{} 0.$$

Indeed, the threshold tends to zero and the Gaussian limit has no mass at zero. Thus the rescaled error is not bounded in probability.

Theorem 2.8 (Multivariate central limit theorem)**.**
Let $X_1, X_2, \dots$ be i.i.d. random vectors in $\mathbb{R}^d$, with d fixed and $\mathbb{E}[\|X\|^2] < \infty$. Then,

$$\sqrt{n}\,(\overline{X}_n - \mathbb{E}[X]) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0, \mathbb{V}(X)).$$

Proof. Let $a \in \mathbb{R}^d$. The real random variables $(a^\top X_i)_{i\in\mathbb{N}}$ are i.i.d. with mean $a^\top \mathbb{E}[X]$ and variance $a^\top \mathbb{V}(X)a$. This variance is finite because Cauchy–Schwarz gives

$$\mathbb{E}[(a^\top X)^2] \le \|a\|^2 \mathbb{E}[\|X\|^2] < \infty.$$

Hence, the CLT gives

$$\sqrt{n}a^\top(\overline{X}_n - \mathbb{E}[X]) = \sqrt{n}\Big(\frac{1}{n}\sum_{i=1}^{n} a^\top X_i - a^\top \mathbb{E}[X]\Big) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0, a^\top \mathbb{V}(X)a).$$

These are the projection laws of $\mathcal{N}(0, \mathbb{V}(X))$. Since the convergence holds for every a, the Cramér–Wold device gives the result. ■

Exercise 2.4 (★)**.**
Let $X_1, X_2, \dots$ be i.i.d. $\mathcal{N}(0,1)$ random variables and set $Q_n := \sum_{i=1}^n X_i^2$. The law of Q_n is called the chi-square law with n degrees of freedom. You may use $\mathbb{E}[X^2] = 1$ and $\mathbb{E}[X^4] = 3$.

(a) Compute $\mathbb{E}[Q_n]$ and $\mathbb{V}(Q_n)$, and show that

$$\frac{Q_n - n}{\sqrt{2n}} \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0,1).$$

(b) Let $\alpha \in (0,1)$ and let $z_{1-\alpha}$ be the standard normal quantile of order $1-\alpha$. Show that

$$\mathbb{P}\left[Q_n \le n + \sqrt{2n}z_{1-\alpha}\right] \xrightarrow[n\to\infty]{} 1 - \alpha.$$

2.2.3 Some statistical applications

The sample mean and sample variance are consistent estimators, but consistency alone does not quantify their uncertainty. The central limit theorem describes the estimation error after multiplication by $\sqrt{n}$. We use it to construct a confidence interval for a mean with unknown variance, then derive an asymptotic distribution for the sample variance.

Throughout this subsection, $X_1, X_2, \ldots$ are i.i.d. real random variables, with mean $\mu := \mathbb{E}[X]$ and variance $\sigma^2 := \mathbb{V}(X) \in (0; \infty)$. Recall that the sample variance is defined by

$$S_n^2 := \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X}_n)^2.$$

Confidence intervals for a mean. The central limit theorem gives

$$\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0,1).$$

Let $\alpha \in (0,1)$, and denote by $z_{1-\alpha/2}$ the quantile of order $1-\alpha/2$ of the standard normal distribution. Thus, if $Z \sim \mathcal{N}(0,1)$, then

$$\mathbb{P}\left[-z_{1-\alpha/2} \le Z \le z_{1-\alpha/2}\right] = 1 - \alpha.$$

If σ were known, we could therefore use the normal approximation to bound the estimation error by $z_{1-\alpha/2}\sigma/\sqrt{n}$, with probability tending to $1-\alpha$. In practice, however, σ is usually unknown too. A natural idea is to replace it by S_n, which we already know is consistent. We must still justify that this replacement preserves the limiting distribution: the denominator is now random, and it is computed from the same observations as the numerator.

Example 2.3 (Asymptotic confidence interval for the mean)**.**
For $\alpha \in (0,1)$, the random interval

$$I_n := \left[\overline{X}_n - z_{1-\alpha/2}\frac{S_n}{\sqrt{n}},\ \overline{X}_n + z_{1-\alpha/2}\frac{S_n}{\sqrt{n}}\right]$$

is an ***asymptotic confidence interval*** for μ with confidence level $1-\alpha$, in the sense that

$$\lim_{n\to\infty} \mathbb{P}\left[\mu \in I_n\right] = 1 - \alpha.$$

Explanation. To establish the coverage, consider the statistic

$$T_n := \frac{\sqrt{n}(\overline{X}_n - \mu)}{S_n},$$

with $T_n = 0$ when $S_n = 0$. By Example 2.1, $S_n^2 \xrightarrow[n\to\infty]{\mathbb{P}} \sigma^2$, so $S_n \xrightarrow[n\to\infty]{\mathbb{P}} \sigma$. Since $\sigma > 0$, Slutsky's theorem gives

$$T_n = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \times \frac{\sigma}{S_n} \xrightarrow[n\to\infty]{\mathcal{L}} Z \times 1 = Z, \qquad Z \sim \mathcal{N}(0,1).$$

Notice that independence between $\overline{X}_n$ and S_n was not needed. The key point is that S_n converges to a constant, so its randomness does not contribute to the limiting distribution of the ratio. This replacement of the unknown standard deviation by an estimate is called ***studentization***. The resulting statistic T_n is an ***asymptotic pivot***: its limiting distribution does not depend on any unknown parameter.

Remark 2.4 (Zero sample variance).
The convention when $S_n = 0$ has no effect on the limit, because

$$\mathbb{P}[S_n = 0] \leq \mathbb{P}\left[|S_n^2 - \sigma^2| > \frac{\sigma^2}{2}\right] \xrightarrow[n\to\infty]{} 0.$$

The ratio σ/S_n is also set to zero on this event.

It remains to turn this convergence in distribution into a statement about the interval. On the event $\{S_n > 0\}$, rearranging the inequalities gives

$$\mu \in I_n \quad \Longleftrightarrow \quad |\overline{X}_n - \mu| \leq z_{1-\alpha/2}\frac{S_n}{\sqrt{n}} \quad \Longleftrightarrow \quad |T_n| \leq z_{1-\alpha/2}.$$

The standard normal distribution puts no probability on the endpoints of $[-z_{1-\alpha/2}, z_{1-\alpha/2}]$. Hence, the Portmanteau lemma gives

$$\mathbb{P}\left[|T_n| \leq z_{1-\alpha/2}\right] \xrightarrow[n\to\infty]{} \mathbb{P}\left[|Z| \leq z_{1-\alpha/2}\right] = 1 - \alpha.$$

The events $\{\mu \in I_n\}$ and $\{|T_n| \leq z_{1-\alpha/2}\}$ can differ only when $S_n = 0$. Their probabilities therefore differ by at most $\mathbb{P}[S_n = 0] \xrightarrow[n\to\infty]{} 0$, proving the coverage statement.

The parameter μ is fixed; it is the interval that varies from one sample to another. The coverage statement says that the probability of this random interval containing μ tends to $1 - \alpha$ as n increases. This is an asymptotic statement, so the coverage need not be exactly $1 - \alpha$ for a given finite sample size. No Gaussian assumption on the observations was used. ♣

Technique. Suppose that $\sqrt{n}(\widehat{\theta}_n - \theta) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0, v)$, with $v > 0$, and that $\widehat{v}_n \xrightarrow[n\to\infty]{\mathbb{P}} v$. Slutsky's theorem allows us to replace v by $\widehat{v}_n$ in the normalization:

$$\frac{\sqrt{n}(\widehat{\theta}_n - \theta)}{\sqrt{\widehat{v}_n}} \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0, 1),$$

with the statistic set to zero when $\widehat{v}_n \leq 0$, an event whose probability tends to zero. Rearranging inequalities involving normal quantiles gives an interval, and the Portmanteau lemma justifies its limiting coverage. Consistency of the variance estimator is enough; no rate of convergence for it is required.

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